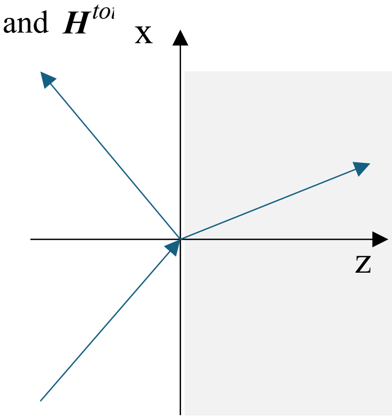
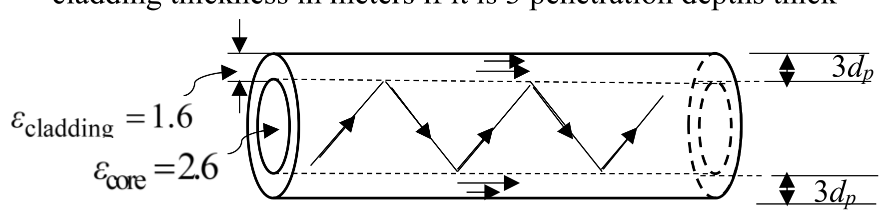
1. A plane wave in air $\boldsymbol{E}^i = 10e^{-j(6x+8z)}\hat{\boldsymbol{y}}$ V/m is incident on a perfectly conducting material at $z = 0$:
   (a) What are its wavelength and frequency
   (b) The time domain expressions for $\boldsymbol{\mathcal{E}}^i(t)$ and $\boldsymbol{\mathcal{H}}^i(t)$
   (c) What is the angle of incidence of the wave
   (d) What is the total frequency domain field in the air region $\boldsymbol{E}^{tot}$ and $\boldsymbol{H}^{tot}$



2. A wave is trapped in an optical fiber as shown in Fig 2 below. Given a fiber with a dielectric constant $\varepsilon_r = 2.6$ is clad in a dielectric with $\varepsilon_r = 1.6$ at 3×10^{14} Hz:
   a. Over what range of angles can the wave propagate in the rod without escaping the rod?
   b. What will be the angle the angle of incidence θ_i if the wave propagates at the minimum angle θ_1 without escaping the rod.
   c. If the angle of incidence of the wave bouncing in the core is 65° what must be the cladding thickness in meters if it is 3 penetration depths thick



3. A light in air shines through a polarized lens toward the surface of a sheet of glass with $\varepsilon_r = 5$. An observer moving vertically above the light source notices that as he can see the reflection of the light in a glass surface at all but a certain distance above the source. If both source and observer are 2 meters from the glass surface (a) how far apart are they? (b) What is the polarization of the light with respect to the plane of the page (the plane of incidence)? (c) What is the index of refraction n_g of the glass? Hint: this is an application of Brewster's angle.